

WEEK 1: PARTICLE WAVE DUALITY

THE DOUBLE SLIT EXPERIMENT

FROM THE QUANTUM UNIVERSE TO QUANTUM TECHNOLOGIES

Week 1 • Spring 2025

1. SIMULATING THE EXPERIMENT

1.1 Classical Behavior: Particles

Imagine you are firing tennis balls at a wall that has two narrow slits in it, labeled A and B, through which balls may pass. The slits are a distance d apart from each other.

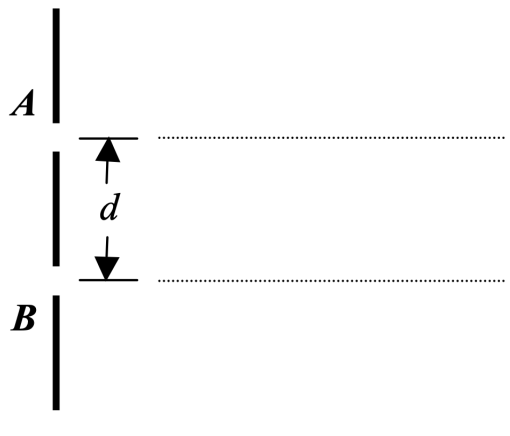


Figure 1: Double Slit Setup with Tennis Balls

1. Draw the pattern of how the balls would hit the wall if you cover slit A. Now what if you cover slit B? If both slits are open, how does this look?

1.2 Classical Behavior: Waves

Now imagine we push water towards our double slitted wall. Let's see how the behavior of waves of water differs from the behavior of the tennis balls. For this experiment, let's look at a simulation:

https://phet.colorado.edu/sims/html/wave-interference/latest/wave-interference_all.html

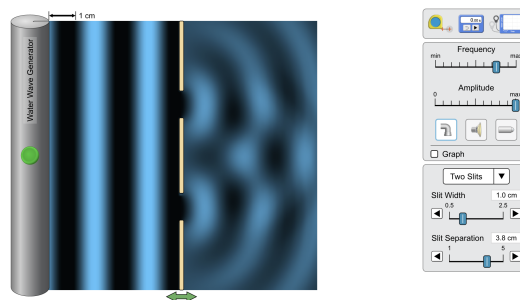


Figure 2: Double Slit Setup with Water Waves

2. First play with the simulation a bit, when there is one slit, draw the pattern you would see when the waves hit the "wall".

3. Now add a second slit. How does interference explain the pattern we see with the water waves?

1.3 Wave-Like Behavior: Light

Next, let's change the settings of our simulation so we are looking at light and not water. We usually think of light as a particle: photons! Choose the option of a single slit and click the option for Screen → Intensity as shown in the figure.

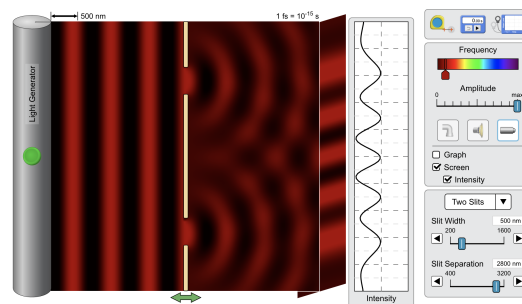


Figure 3: Double Slit Setup with Light

4. First play with the simulation a bit, when there is one slit, draw the pattern of intensity shown by the plot. Is there one place where we are most likely to see light? Is light acting more like the tennis balls or the water waves?

5. Now add a second slit. Draw the pattern we see on the screen, and the intensity. Is light acting more like the tennis balls or the water waves?

1.4 Wave-Like Behavior: Matter

Not only do we see light acting like a wave, we see electrons and even buckyballs (balls of carbon atoms) displaying the same diffraction pattern when we do the double slit experiment. So if matter can behave like waves...and we are made of matter...then so can we! Let's calculate our wavelengths. We can use the de Broglie wavelength λ to calculate.

The de Broglie wavelength is Planck's constant divided by momentum. Momentum can be written as mass times velocity. In order for the de Broglie wavelength to be big, a really tiny momentum is needed. We humans are wayyyy too massive to have large wavelengths, so we behave like particles, not waves.

$$\lambda = \frac{h}{p} \rightarrow \lambda = \frac{h}{mv}$$

6. Calculate your de Broglie wavelength (you need your mass in kilograms) at a speed of 1.2 meters per second (the average walking speed).

7. Usain Bolt is 94 kg, and his top speed was 12.33 m/s (27.33 miles per hour!). Calculate what his wavelength would be during this sprint.